SOHAN DAS

Robotics Engineer · Embedded Systems · Control & Autonomous Systems

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LINKS

GitHub
github.com/sohandas
LinkedIn
linkedin.com/in/sohandas
CPD Portfolio
sohandas.cpd.link
Publication DOI
10.13140/RG.2.2.
33678.54082

LANGUAGES

English
Proficient (C1+)
Bengali
Native
Hindi
Fluent (C1)
Japanese
Basic (A1–A2)
German
Beginner (learning)

CORE SKILLS

Robotics & Control
ROS2 · MATLAB/Simulink · Control Systems · Sensor Fusion · Computer Vision · SLAM · MuJoCo · Gazebo/Rviz · RoboDK · FANUC · Cyberbotics

Programming

PROFESSIONAL PROFILE

Robotics engineer currently pursuing an MSc in Robotics at the University of Manchester, with a strong foundation in control systems, sensor fusion, and embedded electronics. Experienced across the full robotics stack — from custom PCB design and microcontroller firmware to high-level autonomous navigation and Human-Robot Interaction (HRI) research. Published researcher with a track record of delivering hands-on engineering projects. Seeking a challenging role in a European technology organisation to contribute to next-generation robotics and intelligent systems.

EDUCATION

MSc Robotics

University of Manchester • *Sept 2025 – Sept 2026 (Expected)* • Manchester, UK

- Specialisation in advanced control systems, sensor fusion, cognitive robotics, and computer vision
- Research focus: Human-Robot Interaction (HRI) and autonomous navigation systems
- Relevant modules: Embedded Systems, Robot Programming, Machine Learning for Robotics

BTech Electrical & Electronics Engineering

VIT University • *Aug 2020 – Jul 2024* • India

- GPA: 8.46 / 10.0 — graduated in top academic quartile
- Final-year thesis resulted in peer-reviewed publication on humanoid robot arm motion planning
- Core modules: Digital Signal Processing, Power Electronics, Microcontroller Programming, Robot Kinematics

PROFESSIONAL EXPERIENCE

Virtual Robotics & Control Engineer

Forage (Industry Simulation) • *Feb 2026 – Mar 2026* • Remote

- Executed industry-standard robotics engineering workflows simulating real-world control system design
- Implemented and debugged embedded control solutions under structured professional constraints
- Optimised software modules for performance and readability, meeting industry code-quality benchmarks

STEM Teaching Volunteer

International Association of Lions Club • *Jul 2024 – Dec 2024* • India

- Taught mathematics and computer science to students from underserved communities
- Adapted complex STEM concepts for diverse learning levels, strengthening communication skills

RPA Developer Intern

Automation Anywhere • *Jun 2022 – Jul 2022* • India

- Designed and deployed robotic process automation workflows to reduce manual enterprise task load
- Gained practical exposure to the full bot development lifecycle and process optimisation techniques

Python · Embedded C · MATLAB · Java/C++ · R · SQL · HTML/CSS/JS

Hardware & EDA
Arduino · Raspberry Pi · STM32 · PCB Design · OrCAD PSpice · Embedded Systems

Dev Tools
Git/GitHub · Linux (Ubuntu) · Nav2 Stack · Automation Anywhere

KEY ENGINEERING PROJECTS

Autonomous Navigation	Full ROS2 Nav2 pipeline — SLAM-based mapping, localisation, and real-time path execution on a Leo-Rover platform
UAV Position Controller	Cascade PID controller with Disturbance Observer for aerial vehicle stabilisation and trajectory tracking
Path Planning (A*)	A* algorithm implementation for optimal robot path planning in grid-based and dynamic 2D environments
EV Battery Management	Designed a BMS for electric vehicles: cell balancing, thermal monitoring, and safety cutoff circuitry
Humanoid Arm Motion	Research project on 7-DOF humanoid arm trajectory planning — published as peer-reviewed paper (see DOI)
Floor Cleaning Robot	Autonomous cleaning robot with ultrasonic obstacle avoidance and systematic coverage path planning

PUBLICATION

Arm Motion Planning of Humanoid Robot

Peer-Reviewed Research • *2024* • VIT University

- Original research on motion planning algorithms for a 7-DOF humanoid robot arm
- Developed collision-free trajectory generation methods validated in simulation
- DOI: 10.13140/RG.2.2.33678.54082

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

ROS2	Robot Operating System 2 — publishers, subscribers, navigation stack, and lifecycle nodes
Embedded C Programming	Memory management, peripherals, interrupts, and RTOS fundamentals
Robot Motion Planning & Control	Trajectory planning, kinematics, dynamics, and motion control algorithms
Supervised Machine Learning	Regression, classification, model evaluation, and ML pipeline design
Robotics with Arduino	Sensors, actuators, motor drivers, and real-time embedded control
Wix Website Builder	Professional web development and deployment certification
Automation Anywhere RPA	Industry internship — robotic process automation platform and bot lifecycle
CPD Portfolio (Active)	Continuous professional development — all skills and certifications verified (sohandas.cpd.link)

I hereby confirm that all information provided in this curriculum vitae is accurate and complete. — Sohan Das, April 2026